

Toward Scalable and Valid Conditional Independence Testing with Spectral Features

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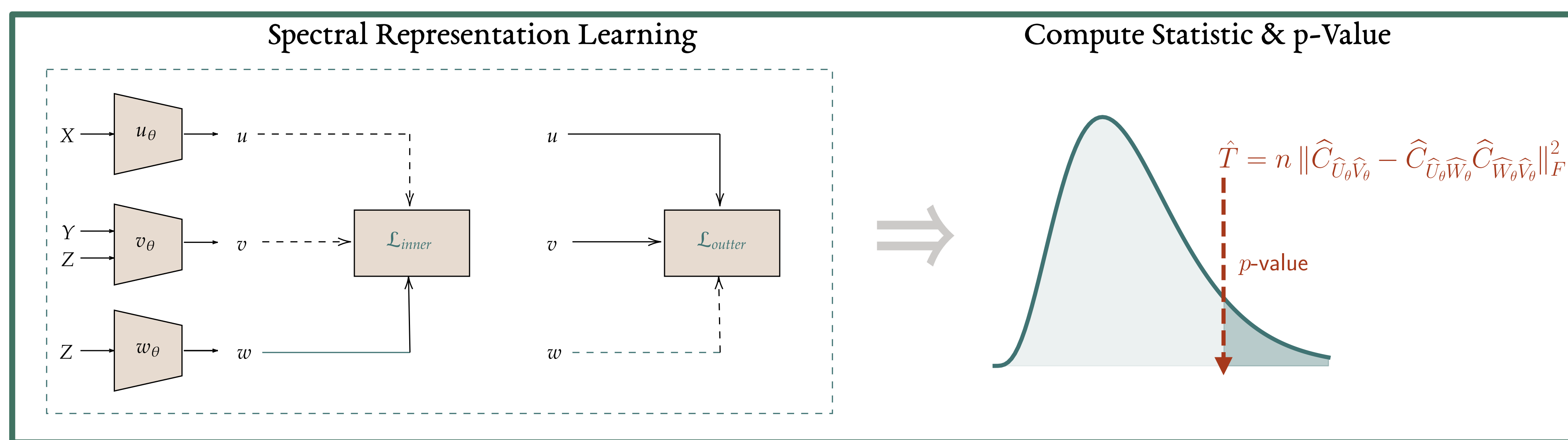
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Our Test (SCIT)

Partial covariance operator:

$$\Sigma_{X\tilde{Y}.Z} = \Sigma_{X\tilde{Y}} - \Sigma_{XZ}\Sigma_{Z\tilde{Y}},$$

Operator formulation of CI:

$$\mathcal{H}_0 : \Sigma_{X\tilde{Y}.Z} = 0 \quad \text{vs.} \quad \mathcal{H}_1 : \|\Sigma_{X\tilde{Y}.Z}\|_F^2 \geq \epsilon_n$$

Losses:

$$\begin{aligned} \hat{\mathcal{L}}_{\text{out}}(\theta) &= \frac{1}{m(m-1)} \sum_{i \neq j}^m \langle u_\theta(x_i), M_\theta v_\theta(\tilde{y}_j) \rangle^2 \\ &\quad - \frac{2}{m-1} \sum_{i=1}^m \langle u_\theta(x_i), M_\theta v_\theta(\tilde{y}_i) \rangle \\ &\quad + \frac{2}{m(m-1)} \sum_{i \neq j}^m \langle u_\theta(x_i), v_\theta(\tilde{y}_j) \rangle \langle w_\theta(z_i), M_\theta w_\theta(z_j) \rangle \\ \hat{\mathcal{L}}_{\text{in}}(\theta) &= \frac{1}{m(m-1)} \sum_{i \neq j}^m \langle w_\theta(z_i), N_\theta w_\theta(z_j) \rangle^2 \\ &\quad - \frac{2}{m(m-1)} \sum_{i \neq j}^m \langle u_\theta(x_i), N_\theta v_\theta(\tilde{y}_j) \rangle \langle w_\theta(z_i), M_\theta w_\theta(z_j) \rangle \end{aligned}$$

Orthonormality (whitening) regularizers:

$$\begin{aligned} \hat{\Omega}_{\text{out}}(\theta) &= \|\hat{C}_{U_\theta U_\theta} - I_d\|_F^2 + \|\hat{C}_{V_\theta V_\theta} - I_d\|_F^2 \\ \hat{\Omega}_{\text{in}}(\theta) &= \|\hat{C}_{W_\theta W_\theta} - I_d\|_F^2. \end{aligned}$$

Algorithm: SCIT

Input: $\{(x_i, y_i, z_i)\}_{i=1}^n$; regularization strength $\gamma > 0$

Learning representations

for $t = 1, \dots, n_{\text{steps}}$ **do**

Training inner model

for $s = 1, \dots, n_{\text{steps_inner}}$ **do**

Sample mini-batch $\mathbf{B} \leftarrow (\mathbf{x}, \mathbf{y}, \mathbf{z})$

$g_{\text{in}} \leftarrow \nabla_\theta [\hat{\mathcal{L}}_{\text{in}}(\theta, \mathbf{B}) + \gamma \hat{\Omega}_{\text{in}}(\theta, \mathbf{B})]$

Update w_θ with g_{in}

Training outer model

Sample mini-batch $\mathbf{B}' \leftarrow (\mathbf{x}, \mathbf{y}, \mathbf{z})$

$g_{\text{out}} \leftarrow \nabla_\theta [\hat{\mathcal{L}}_{\text{out}}(\theta, \mathbf{B}') + \gamma \hat{\Omega}_{\text{out}}(\theta, \mathbf{B}')]$

Update u_θ, v_θ with g_{out}

Computing test statistic and p-value

Compute test statistic $\hat{T} \leftarrow n \|\hat{C}_{\hat{U}_\theta \hat{V}_\theta} - \hat{C}_{\hat{U}_\theta \hat{W}_\theta} \hat{C}_{\hat{W}_\theta \hat{V}_\theta}\|_F^2$

Output: p-value obtained by locating $\hat{T}$ on $\chi^2(d^2)$

Statistical Analysis

Representation learning error:

$$\mathcal{E}_m = \max \left\{ \|C_{\hat{U}_\theta \hat{U}_\theta} - I_d\|, \|C_{\hat{V}_\theta \hat{V}_\theta} - I_d\|, \|C_{\hat{W}_\theta \hat{W}_\theta} - I_d\|, \|\Sigma_{X\tilde{Y}.Z} - U_\theta M_\theta V_\theta^*\| \right\}$$

Theorem: Type I Error

Let $\|\hat{u}_\theta(X)\|_2, \|\hat{v}_\theta(Y, Z)\|_2, \|\hat{w}_\theta(Z)\|_2$ be sub-Gaussian random variables, then under the null

$$\hat{T} \rightarrow^D \sum_{i,j}^d \lambda_i(\hat{u}) \lambda_j(\hat{v}) Z_{i,j}^2 \quad \text{as } n \rightarrow \infty$$

where $Z_{i,j} \sim \mathcal{N}(0, 1)$, $\lambda_i(\hat{u}), \lambda_j(\hat{v})$ are the eigenvalues of $\hat{C}_{\hat{U}_\theta \hat{U}_\theta}, \hat{C}_{\hat{V}_\theta \hat{V}_\theta}$, and d is the latent dimension of the networks. Furthermore, if we assume $\mathcal{E}_m \rightarrow 0$ as $m \rightarrow \infty$, then

$$\hat{T} \rightarrow^D \chi^2(d^2) \quad \text{as } m, n \rightarrow \infty$$

Outline

Problem:

- Given IID sample (X_i, Y_i, Z_i) from P_{XYZ} , decide

$$\mathcal{H}_0 : X \perp\!\!\!\perp Y \mid Z \quad \text{vs.} \quad \mathcal{H}_1 : X \not\perp\!\!\!\perp Y \mid Z$$

Applications:

- Causal discovery, causal graph validation, etc.

Challenge:

- CI is frequently untestable in important settings without additional assumptions [1]
- Existing methods often make restrictive assumptions

Operator formulation of CI testing:

- More flexible nonparametric formulation through partial covariance operator
- This approach underlies classical kernel methods [2]
- Limitations of kernels:** lack of adaptivity, slow convergence and poor scalability [3]

Bridging kernels with representation learning:

- Spectral contrastive learning recently has shown promise in nonparametric inference [4, 5]
- We provide a contrastive algorithm to learn leading spectral features of partial covariance operator
- Our theoretical analysis links representation quality to test performance using a simple test statistic

Testing Independence by Spectral Features

Gaussian setting:

- Covariance $C_{XY} = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])^\top]$
- $X \perp\!\!\!\perp Y \iff C_{XY} = 0$

Detecting nonlinear dependencies:

- $f : \mathcal{X} \rightarrow \mathbb{R}, g : \mathcal{Y} \rightarrow \mathbb{R}$
- $X \perp\!\!\!\perp Y \iff \sup_{f \in \mathcal{F}, g \in \mathcal{G}} \text{cov}(f(X), g(Y)) = 0$

Covariance operator:

- Put coordinates f_i, g_j on $L^2(X), L^2(Y)$
- Covariance operator (infinite matrix) captures all dependencies

$$\langle f_i, \Sigma_{XY} g_j \rangle = \text{cov}(f_i(X), g_j(Y))$$

- Hence, $X \perp\!\!\!\perp Y \iff \|\Sigma_{XY}\|_{\text{HS}} = 0$ (HSIC)

Singular Value Decomposition:

- Σ_{XY} can be written as $\Sigma_{XY} = \sum_{i=1}^\infty \sigma_i u_i v_i^\top$
- Eckart–Young–Mirsky theorem:** rank- d truncated SVD is the best rank- d approximation of any matrix / compact operator

Spectral Contrastive Learning:

- How to learn truncated SVD of covariance operator?
- In practice, it boils down to spectral contrastive learning

$$\begin{aligned} \mathcal{L}_{\text{unconditional}}(\theta) &= -\frac{2}{m} \sum_{i=1}^m \langle u_\theta(x_i), \sigma_\theta \odot v_\theta(y_i) \rangle^2 \\ &\quad + \frac{1}{m(m-1)} \sum_{i \neq j}^m \langle u_\theta(x_i), \sigma_\theta \odot v_\theta(y_j) \rangle \end{aligned}$$

with u_θ, v_θ neural nets and σ_θ weights representing the singular functions and values to be learned

Testing:

- Compute $\hat{S} = \|\hat{C}_{\hat{U}_\theta \hat{V}_\theta}\|_F^2$ and associated p-value against null distribution (asymptotically chi-squared)

Theorem: Power

Let $\|\hat{u}_\theta(X)\|_2, \|\hat{v}_\theta(Y, Z)\|_2, \|\hat{w}_\theta(Z)\|_2$ be K-sub-Gaussian random variables, $\mathcal{E}_m \rightarrow 0$ as $m \rightarrow \infty$, and $\epsilon_n \gtrsim K^2 \frac{d}{\sqrt{n}}$, then for the alternative hypothesis,

$$\mathbb{P}_{\mathcal{H}_1} \left(n \|\hat{U}_\theta^\top \hat{V}_\theta\|_F^2 > q_{1-\alpha}(\chi^2(d^2)) \right) \rightarrow 1,$$

as $m, n \rightarrow \infty$, where d is the latent dimension of networks

Experiments

Post-nonlinear Model

- Baselines: KCI [2], GCIT [6]
- Type 1 Error (Figure 1)
- Power (Figure 2)

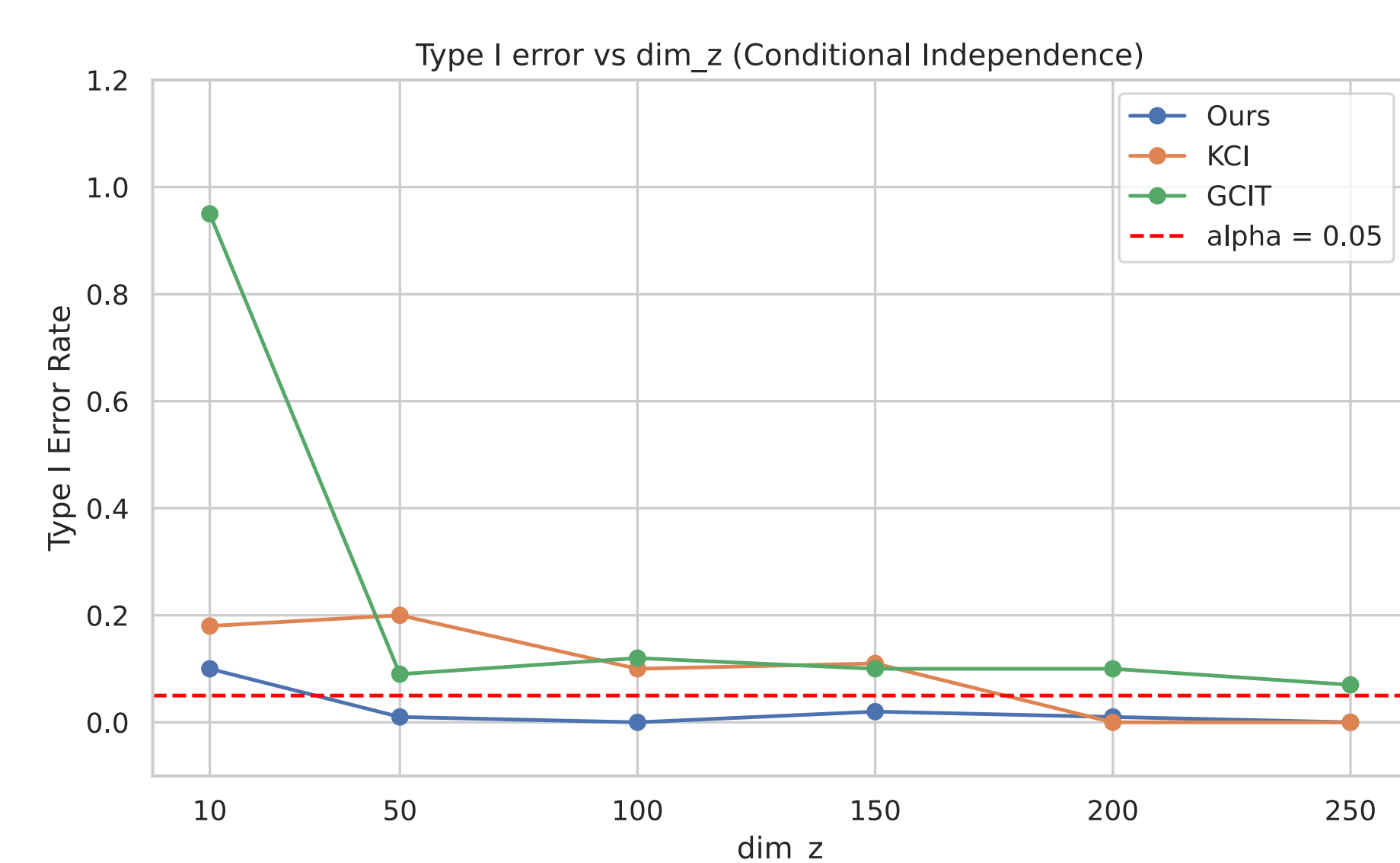


Figure 1: Type 1 Error

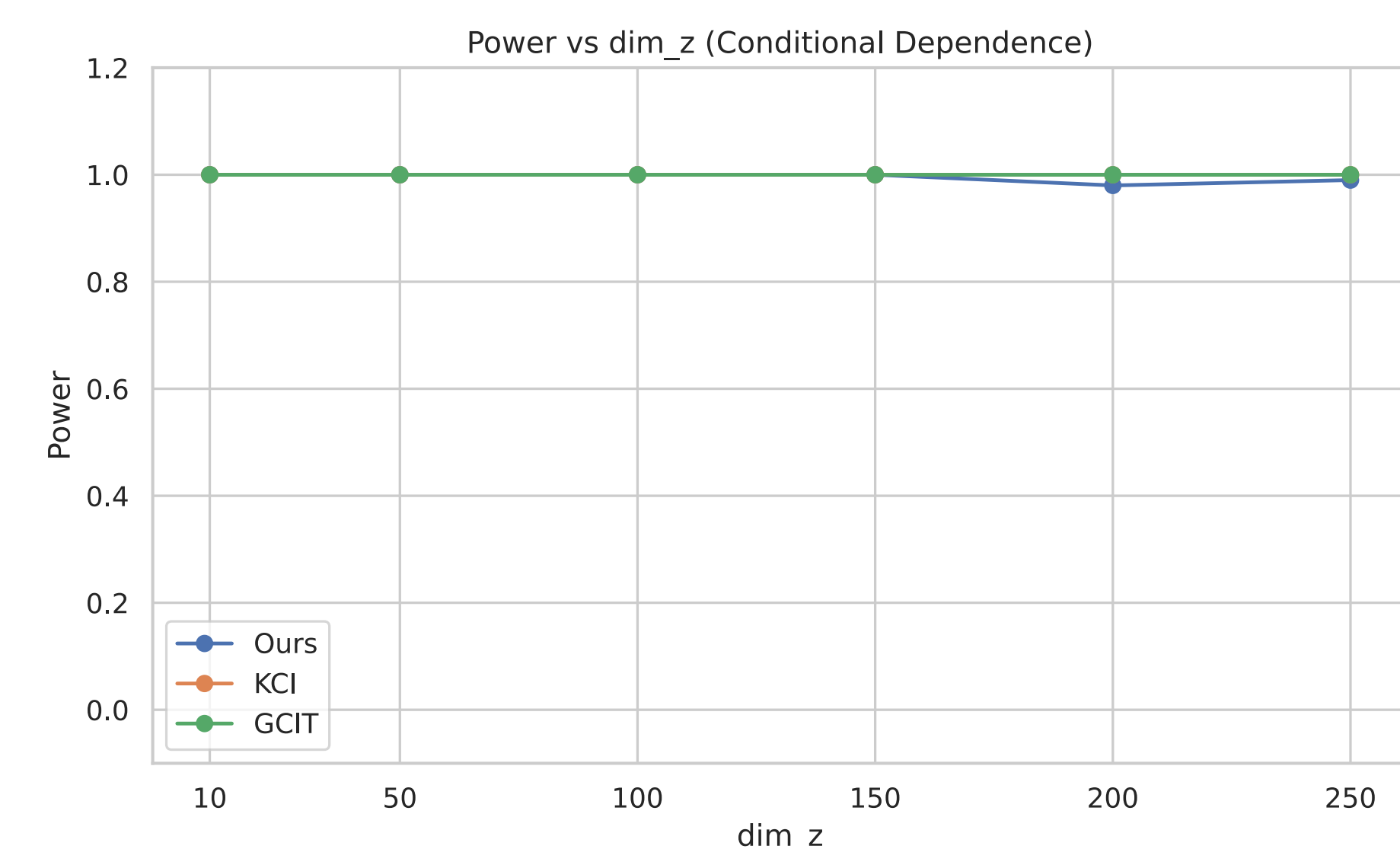


Figure 2: Power

Challenge: Lack of real-world benchmarks (data) in SoTA; how can we do it better?

Project



References

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